بسم الله الرحمن الرحيم

CS 458 - Data Mining (DM)

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Lecture 12: Frequent Pattern Mining

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Mining Frequent Patterns

Itemset Mining, Association and Correlation Analysis

Chapter 6 of the Text Book

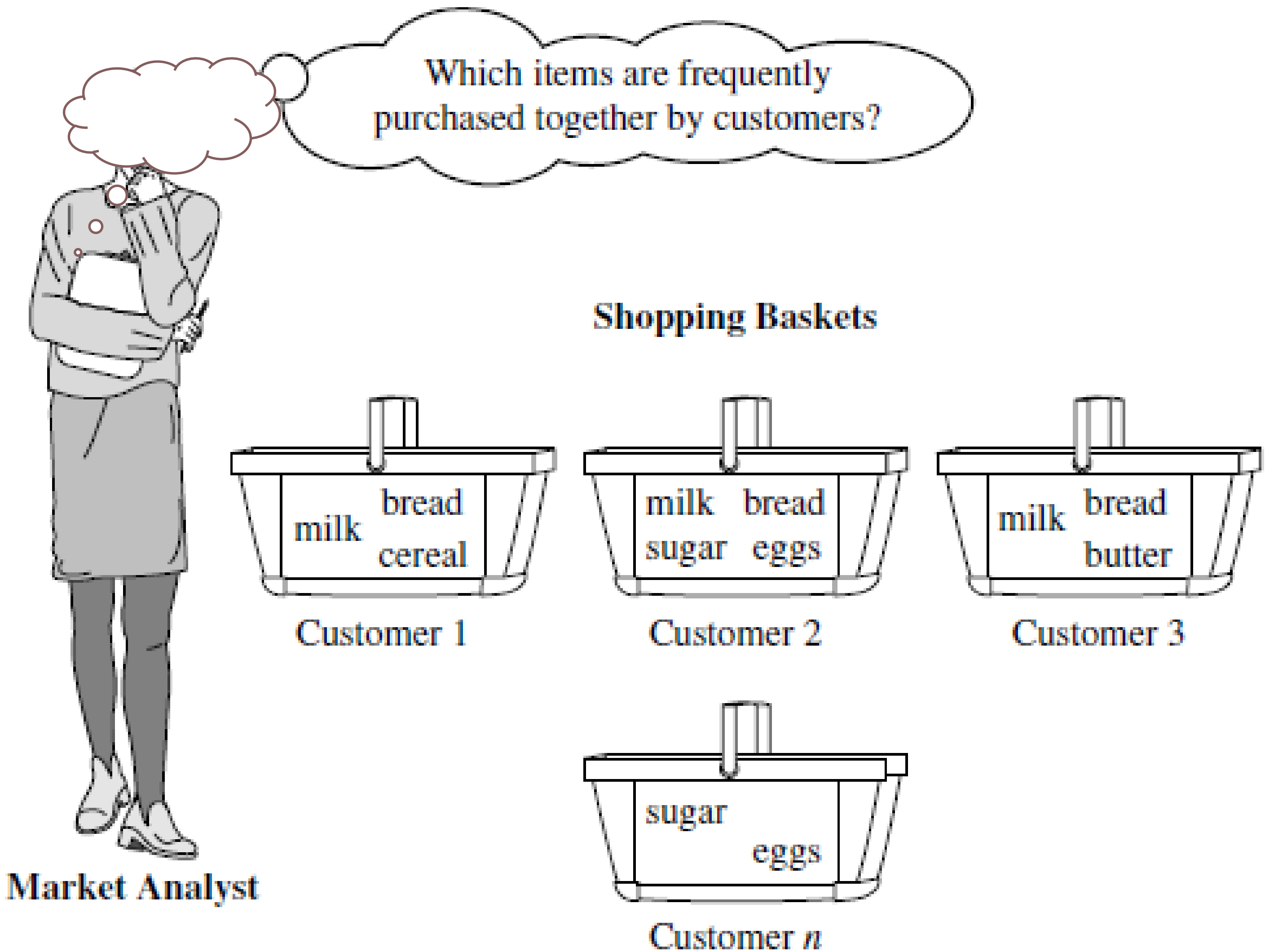


Figure 6.1 Market basket analysis.

Chapter 6: Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

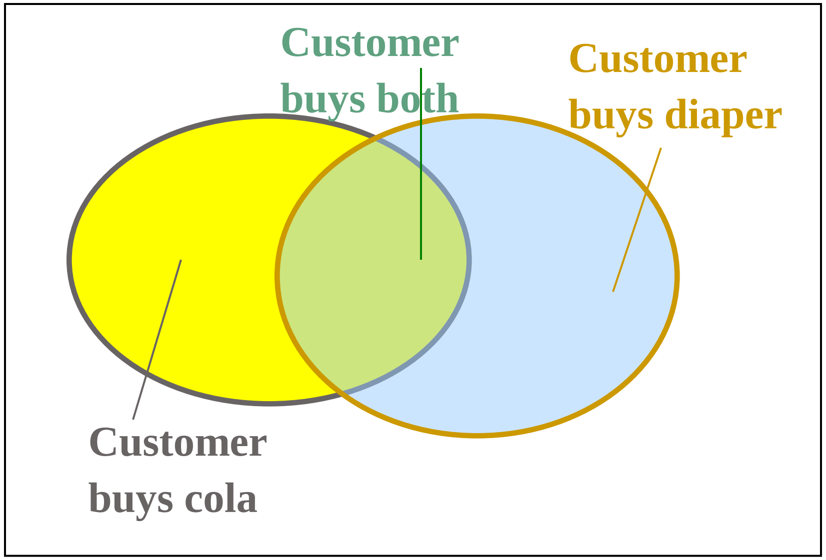
- Basic Concepts
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary

What Is Frequent Pattern Analysis?

- Frequent pattern: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [AIS93] in the context of frequent itemsets and association rule mining
- Motivation: Finding inherent regularities in data
  - What products were often purchased together?— cola and diapers?!
  - What are the subsequent purchases after buying a PC?
  - What kinds of DNA are sensitive to this new drug?
- Applications
  - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis.

Basic Concepts: Frequent Patterns

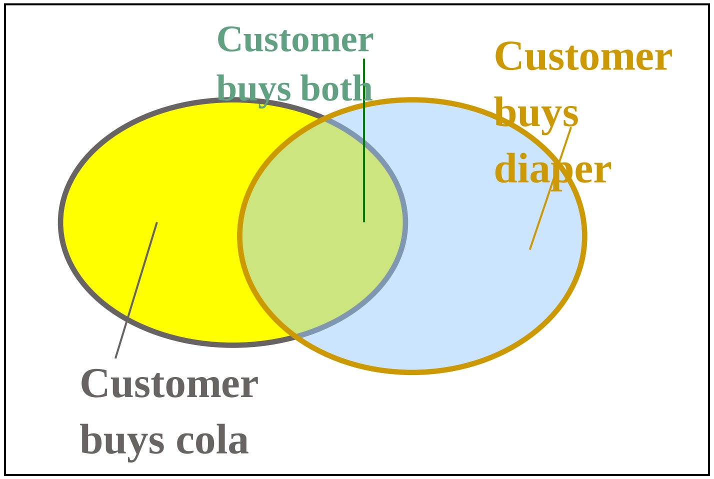
Tid	Items bought
10	cola, Nuts, Diaper
20	cola, Coffee, Diaper
30	cola, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- itemset: A set of one or more items
- k-itemset $X = \{x_1, \ldots, x_k\}$
- *(absolute) support*, or, *support count* of X: Frequency or occurrence of an itemset X
- *(relative) support*, *s*, is the fraction of transactions that contains X (i.e., the probability that a transaction contains X)
- An itemset X is *frequent* if X's support is no less than a *minsup* threshold

Basic Concepts: Association Rules

Tid	Items bought
10	cola, Nuts, Diaper
20	cola, Coffee, Diaper
30	cola, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- Find all the rules $X \rightarrow Y$ with minimum support and confidence
  - support, s, probability that a transaction contains $X \cup Y$
  - confidence, c, conditional probability that a transaction having X also contains Y

Let minsup = 50%, minconf = 50%

Freq. Pat.: cola:3, Nuts:3, Diaper:4, Eggs:3, {cola, Diaper}:3

- Association rules: (many more!)
  - *cola* → *Diaper* (60%, 100%)
  - *Diaper* → *cola* (60%, 75%)

Closed Patterns and Max-Patterns

- A long pattern contains a combinatorial number of sub-patterns, e.g., $\{a_1, \ldots, a_{100}\}$ contains $\binom{100}{1} + \binom{100}{2} + \ldots + \binom{100}{100} = 2^{100} - 1 = 1.27 * 10^{30}$ sub-patterns!
- Solution: *Mine closed patterns and max-patterns instead*
- An itemset X is closed if X is *frequent* and there exists *no super-pattern* Y $\supset$ X, *with the same support* as X
- An itemset X is a max-pattern if X is frequent and there exists no frequent super-pattern Y $\supset$ X
- Closed pattern is a lossless compression of freq. patterns
  - Reducing the # of patterns and rules

Generate Rules using *Apriori Algorithm*, Consider the values as Support = 50% and Confidence =75%.

ID	Itemsets
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

ID	Itemsets
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
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4	Bread, Milk, Juice
5	Butter, Milk, Juice

1-itemset Support-Count L1

Item	Count	Support
Bread	4	80%
Butter	3	60%
Milk	4	80%
Juice	3	60%

1-itemset Support-Count L1

Item	Count	Support
Bread	4	80%
Butter	3	60%
Jam	1	20%
Milk	4	80%
Juice	3	60%
Curd	1	20%

1-itemset Support-Count L2

Item	Count	Support
Bread, Butter	2	40%
Bread, Milk	3	60%
Bread, Juice	2	40%
Butter, Milk	3	60%
Butter, Juice	1	20%
Milk, Juice	2	40%

ID	Itemsets
1	Bread, Butter, Jam, Milk
2	Bread, Butter, Milk
3	Bread, Juice, Curd
4	Bread, Milk, Juice
5	Butter, Milk, Juice

1-itemset Support-Count L2

Item	Count	Support
Bread, Milk	3	60%
Butter, Milk	3	60%

1-itemset Support-Count L3

Item	Count	Support
Bread, Milk, Butter	2	40%

1-itemset Support-Count L2

Item	Count	Support
Bread, Butter	2	40%
Bread, Milk	3	60%
Bread, Juice	2	40%
Butter, Milk	3	60%
Butter, Juice	1	20%
Milk, Juice	2	40%

{Bread, Milk} :

Confi of (Bread → Milk) = 60% / 80% = 75%

Confi of (Milk → Bread) = 75%

{Butter, Milk} :

Confi of (Butter → Milk) = 60% / 60% = 100%

Confi of (Milk → Butter) = 75%

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Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach
- Improving the Efficiency of Apriori
- FPGrowth: A Frequent Pattern-Growth Approach
- ECLAT: Frequent Pattern Mining with Vertical Data Format

The Downward Closure Property and Scalable Mining Methods

- The downward closure property of frequent patterns
  - Any subset of a frequent itemset must be frequent
  - If {**cola, diaper, nuts**} is frequent, so is {**cola, diaper**}
  - i.e., every transaction having {cola, diaper, nuts} also contains {cola, diaper}

Apriori: A Candidate Generation & Test Approach

- Apriori pruning principle: If there is any itemset which is infrequent, its superset should not be generated/tested!
- Method:
  - Initially, scan DB once to get frequent 1-itemset
  - Generate length (k+1) candidate itemsets from length k frequent itemsets
  - Prune all itemsets having any infrequent subset
  - Test the candidates against DB
  - Terminate when no frequent or candidate set can be generated

The Apriori Algorithm—An Example

$Sup_{min} = 2$

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

1^{st} scan

C_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

C_2

Itemset
{A, B}
{A, C}
{A, E}
{B, C}
{B, E}
{C, E}

2^{nd} scan

C_2

Itemset	sup
{A, B}	1
{A, C}	2
{A, E}	1
{B, C}	2
{B, E}	3
{C, E}	2

L_2

Itemset	sup
{A, C}	2
{B, C}	2
{B, E}	3
{C, E}	2

C_3

Itemset
{B, C, E}

3^{rd} scan

L_3

Itemset	sup
{B, C, E}	2

The Apriori Algorithm (Pseudo-Code)

C_k: Candidate itemset of size k

L_k : frequent itemset of size k

L_1 = {frequent items};

for (k = 1; L_k !=$\varnothing$; k++) **do begin**

C_{k+1} = candidates generated from L_k;

for each transaction t in database do

increment the count of all candidates in C_{k+1} that are contained in t

L_{k+1} = candidates in C_{k+1} with min_support

end

return $\cup_k L_k$;

Implementation of Apriori

- How to generate candidates?
  - Step 1: self-joining L_k
  - Step 2: pruning
- Example of Candidate-generation
  - $L_3=\{abc, abd, acd, ace, bcd\}$
  - Self-joining: L_3*L_3
    - *abcd* from *abc* and *abd*
    - *acde* from *acd* and *ace*
  - Pruning:
    - *acde* is removed because *ade* is not in L_3
  - $C_4 = \{abcd\}$

Exercise

Table 6.1 Transactional Data for an *AllElectronics* Branch

TID	*List of item_IDs*
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

illustrate the Apriori algorithm for finding frequent itemsets in *D*.

Let minimum support = 2

Scan D for count of each candidate $\rightarrow$

C_1

Itemset	Sup. count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

Compare candidate support count with minimum support count $\rightarrow$

L_1

Itemset	Sup. count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

Generate C_2 candidates from L_1 $\rightarrow$

C_2

Itemset
{I1, I2}
{I1, I3}
{I1, I4}
{I1, I5}
{I2, I3}
{I2, I4}
{I2, I5}
{I3, I4}
{I3, I5}
{I4, I5}

Scan D for count of each candidate $\rightarrow$

C_2

Itemset	Sup. count
{I1, I2}	4
{I1, I3}	4
{I1, I4}	1
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0
{I3, I5}	1
{I4, I5}	0

Compare candidate support count with minimum support count $\rightarrow$

L_2

Itemset	Sup. count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

Generate C_3 candidates from L_2 $\rightarrow$

C_3

Itemset
{I1, I2, I3}
{I1, I2, I5}

Scan D for count of each candidate $\rightarrow$

C_3

Itemset	Sup. count
{I1, I2, I3}	2
{I1, I2, I5}	2

Compare candidate support count with minimum support count $\rightarrow$

L_3

Itemset	Sup. count
{I1, I2, I3}	2
{I1, I2, I5}	2

Generation of the candidate itemsets and frequent itemsets, where the minimum support count is 2.

Exercise...

- What about L4?

Generating Association Rules from Frequent Itemsets

- For each frequent itemset l, generate all nonempty subsets of l.
- For every nonempty subset s of l, output the rule
  - s → l - s

Generating association rules. Let's try an example based on the transactional data for *AllElectronics* shown before in Table 6.1. The data contain frequent itemset $X = \{I1, I2, I5\}$. What are the association rules that can be generated from X? The nonempty subsets of X are $\{I1, I2\}$, $\{I1, I5\}$, $\{I2, I5\}$, $\{I1\}$, $\{I2\}$, and $\{I5\}$. The resulting association rules are as shown below, each listed with its confidence:

$$
\begin{array}{ll}
\{I1, I2\} \Rightarrow I5, & \textit{confidence} = 2/4 = 50\% \\
\{I1, I5\} \Rightarrow I2, & \textit{confidence} = 2/2 = 100\% \\
\{I2, I5\} \Rightarrow I1, & \textit{confidence} = 2/2 = 100\% \\
I1 \Rightarrow \{I2, I5\}, & \textit{confidence} = 2/6 = 33\% \\
I2 \Rightarrow \{I1, I5\}, & \textit{confidence} = 2/7 = 29\% \\
I5 \Rightarrow \{I1, I2\}, & \textit{confidence} = 2/2 = 100\%
\end{array}
$$

If the minimum confidence threshold is, say, 70%, then only the second, third, and last rules are output, because these are the only ones generated that are strong. Note that, unlike conventional classification rules, association rules can contain more than one conjunct in the right side of the rule. ■

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L_1 = {frequent items};

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